

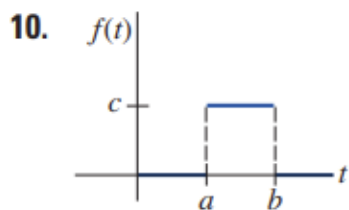
ENGR 311 — Problems

Section 4.1 · ed. 5 p.213

In Problems 1–18, use Definition 4.1.1 to find $\mathcal{L}\{f(t)\}$.

3. $f(t) = \begin{cases} t, & 0 \leq t < 1 \\ 1, & t \geq 1 \end{cases}$

6. $f(t) = \begin{cases} \sin t, & 0 \leq t < \pi/2 \\ 0, & t \geq \pi/2 \end{cases}$



14. $f(t) = t^2 e^{-2t}$

Section 4.2 · ed. 5 p.221

In Problems 1–30, use Theorem 4.2.1 to find the given inverse

5. $\mathcal{L}^{-1}\left\{\frac{(s+1)^3}{s^4}\right\}$

6. $\mathcal{L}^{-1}\left\{\frac{(s+2)^2}{s^3}\right\}$

13. $\mathcal{L}^{-1}\left\{\frac{4s}{4s^2+1}\right\}$

16. $\mathcal{L}^{-1}\left\{\frac{s+1}{s^2+2}\right\}$

In Problems 31–40, use the Laplace transform to solve the given initial-value problem.

36. $y'' - 4y' = 6e^{3t} - 3e^{-t}, \quad y(0) = 1, \quad y'(0) = -1$

40. $y'' + 9y = e^t, \quad y(0) = 0, \quad y'(0) = 0$

Section 4.3 · ed. 5 p.229, p.230

In Problems 1–20, find either $F(s)$ or $f(t)$, as indicated.

5. $\mathcal{L}\{t(e^t + e^{2t})^2\}$

12. $\mathcal{L}^{-1}\left\{\frac{1}{(s-1)^4}\right\}$

In Problems 21–30, use the Laplace transform

22. $y' - y = 1 + te^t, \quad y(0) = 0$

24. $y'' - 4y' + 4y = t^3 e^{2t}, \quad y(0) = 0, \quad y'(0) = 0$

In Problems 37–48, find either $F(s)$ or $f(t)$, as indicated.

41. $\mathcal{L}\{\cos 2t \cdot \mathcal{U}(t - \pi)\}$

45. $\mathcal{L}^{-1}\left\{\frac{e^{-\pi s}}{s^2 + 1}\right\}$

Section 4.4 · ed. 5 p.240, p.241

In Problems 1–8, use Theorem 4.4.1 to evaluate the given Laplace transform.

4. $\mathcal{L}\{t \sinh 3t\}$

7. $\mathcal{L}\{te^{2t} \sin 6t\}$

8. $\mathcal{L}\{te^{-3t} \cos 3t\}$

In Problems 9–14, use the Laplace transform to solve the given

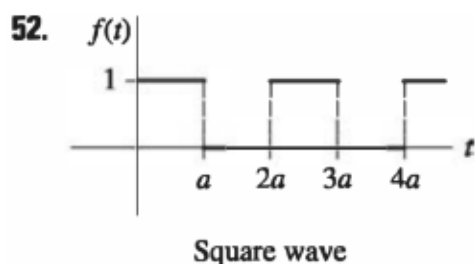
10. $y' - y = te^t \sin t, \quad y(0) = 0$

In Problems 37–46, use the Laplace transform to solve the given integral equation or integrodifferential equation.

45. $y'(t) = 1 - \sin t - \int_0^t y(\tau) d\tau, \quad y(0) = 0$

46. $\frac{dy}{dt} + 6y(t) + 9 \int_0^t y(\tau) d\tau = 1, \quad y(0) = 0$

In Problems 51–56, use Theorem 4.4.3 to find the Laplace transform of the given periodic function.



Section 4.5 · ed. 5 p.245

In Problems 1–12, use the Laplace transform to solve the given differential equation subject to the indicated initial conditions.

3. $y'' + y = \delta(t - 2\pi), \quad y(0) = 0, y'(0) = 1$

9. $y'' + 4y' + 5y = \delta(t - 2\pi), \quad y(0) = 0, y'(0) = 0$

12. $y'' - 7y' + 6y = e^t + \delta(t - 2) + \delta(t - 4), \quad y(0) = 0, y'(0) = 0$

Section 4.6 · ed. 5 p.248, p.249

In Problems 1–12, use the Laplace transform to solve the given system of differential equations.

4. $\frac{dx}{dt} + 3x + \frac{dy}{dt} = 1$

$$\frac{dx}{dt} - x + \frac{dy}{dt} - y = e^t$$

$$x(0) = 0, y(0) = 0$$

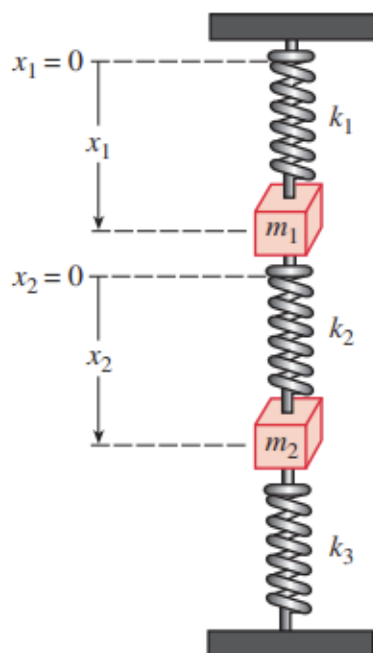
10. $\frac{dx}{dt} - 4x + \frac{d^3y}{dt^3} = 6 \sin t$

$$\frac{dx}{dt} + 2x - 2\frac{d^3y}{dt^3} = 0$$

$$x(0) = 0, y(0) = 0,$$

$$y'(0) = 0, y''(0) = 0$$

14. Derive the system of differential equations describing the straight-line vertical motion of the coupled springs shown in equilibrium in **FIGURE 4.6.4**. Use the Laplace transform to solve the system when $k_1 = 1, k_2 = 1, k_3 = 1, m_1 = 1, m_2 = 1$ and $x_1(0) = 0, x_1'(0) = -1, x_2(0) = 0, x_2'(0) = 1$.



Section 12.1 · ed. 5 p.659

In Problems 1–6, show that the given functions are orthogonal on the indicated interval.

3. $f_1(x) = e^x, f_2(x) = xe^{-x} - e^{-x}; [0, 2]$
6. $f_1(x) = e^x, f_2(x) = \sin x; [\pi/4, 5\pi/4]$

In Problems 7–12, show that the given set of functions is orthogonal on the indicated interval. Find the norm of each function in the set.

10. $\left\{ \sin \frac{n\pi}{p}x \right\}, n = 1, 2, 3, \dots; [0, p]$
18. From Problem 1 we know that $f_1(x) = x$ and $f_2(x) = x^2$ are orthogonal on $[-2, 2]$. Find constants c_1 and c_2 such that $f_3(x) = x + c_1x^2 + c_2x^3$ is orthogonal to both f_1 and f_2 on the same interval.

21. A real-valued function is said to be **periodic** with period $T \neq 0$ if $f(x + T) = f(x)$ for all x in the domain of f . If T is the smallest positive value for which $f(x + T) = f(x)$ holds, then T is called the **fundamental period** of f . In Problems 21–26, determine the fundamental period T of the given function.

(a) $f(x) = \cos 2\pi x$

(b) $f(x) = \sin \frac{4}{L} x$

(c) $f(x) = \sin x + \sin 2x$

(d) $f(x) = \sin 2x + \cos 4x$

(e) $f(x) = \sin 3x + \cos 2x$

(f) $f(x) = A_0 + \sum_{n=1}^{\infty} \left(A_n \cos \frac{n\pi}{p} x + B_n \sin \frac{n\pi}{p} x \right), A_n \text{ and } B_n$
depend only on n

Section 12.2 · ed. 5 p.664

In Problems 1–16, find the Fourier series of f on the given

$$10. f(x) = \begin{cases} 0, & -\pi/2 < x < 0 \\ \cos x, & 0 \leq x < \pi/2 \end{cases}$$

$$12. f(x) = \begin{cases} 0, & -2 < x < 0 \\ x, & 0 \leq x < 1 \\ 1, & 1 \leq x < 2 \end{cases}$$

$$16. f(x) = \begin{cases} 0, & -\pi < x < 0 \\ e^x - 1, & 0 \leq x < \pi \end{cases}$$

Section 12.3 · ed. 5 p.669, p.670

In Problems 1–10, determine whether the given function is even, odd, or neither.

$$4. f(x) = x^3 - 4x$$

$$6. f(x) = e^x - e^{-x}$$

In Problems 11–24, expand the given function in an appropriate cosine or sine series.

$$12. f(x) = \begin{cases} 1, & -2 < x < -1 \\ 0, & -1 < x < 1 \\ 1, & 1 < x < 2 \end{cases}$$

In Problems 25–34, find the half-range cosine and sine expansions of the given function.

$$30. f(x) = \begin{cases} 0, & 0 < x < \pi \\ x - \pi, & \pi \leq x < 2\pi \end{cases}$$

$$33. f(x) = x^2 + x, \quad 0 < x < 1$$

Section 13.1 · ed. 5 p.692

In Problems 1–16, use separation of variables to find, if possible, product solutions for the given partial differential equation.

$$3. u_x + u_y = u$$

$$8. y \frac{\partial^2 u}{\partial x \partial y} + u = 0$$

In Problems 17–26, classify the given partial differential equation as hyperbolic, parabolic, or elliptic.

24. $\frac{\partial^2 u}{\partial x^2} + \frac{\partial^2 u}{\partial y^2} = u$

25. $a^2 \frac{\partial^2 u}{\partial x^2} = \frac{\partial^2 u}{\partial t^2}$

26. $k \frac{\partial^2 u}{\partial x^2} = \frac{\partial u}{\partial t}, k > 0$

Section 13.2 · ed. 5 p.697

In Problems 1–6, a rod of length L coincides with the interval $[0, L]$ on the x -axis. Set up the boundary-value problem for the temperature $u(x, t)$.

3. The left end is held at temperature 100° , and there is heat transfer from the right end into the surrounding medium at temperature zero. The initial temperature is $f(x)$ throughout.

In Problems 7–10, a string of length L coincides with the interval $[0, L]$ on the x -axis. Set up the boundary-value problem for the displacement $u(x, t)$.

10. The ends are secured to the x -axis, and the string is initially at rest on that axis. An external vertical force proportional to the horizontal distance from the left end acts on the string for $t > 0$.

In Problems 11 and 12, set up the boundary-value problem for the steady-state temperature $u(x, y)$.

12. A semi-infinite plate coincides with the region defined by $0 \leq x \leq \pi, y \geq 0$. The left end is held at temperature e^{-y} , and the right end is held at temperature 100° for $0 < y \leq 1$ and temperature zero for $y > 1$. The bottom of the plate is held at temperature $f(x)$.

Section 13.3 · ed. 5 p.699

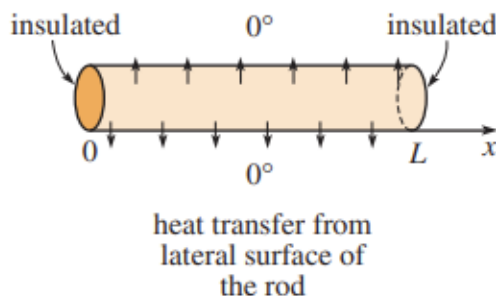
In Problems 1 and 2, solve the heat equation (1) subject to the given conditions. Assume a rod of length L .

2. $u(0, t) = 0, \quad u(L, t) = 0$
 $u(x, 0) = x(L - x)$

5. Suppose heat is lost from the lateral surface of a thin rod of length L into a surrounding medium at temperature zero. If the linear law of heat transfer applies, then the heat equation

$$k \frac{\partial^2 u}{\partial x^2} - hu = \frac{\partial u}{\partial t}, \quad 0 < x < L, \quad t > 0,$$

h a constant. Find the temperature $u(x, t)$ if the initial temperature is $f(x)$ throughout and the ends $x = 0$ and $x = L$ are insulated.



Section 13.4 · ed. 5 p.703, p.704

In Problems 1–6, solve the wave equation (1) subject to the given conditions.

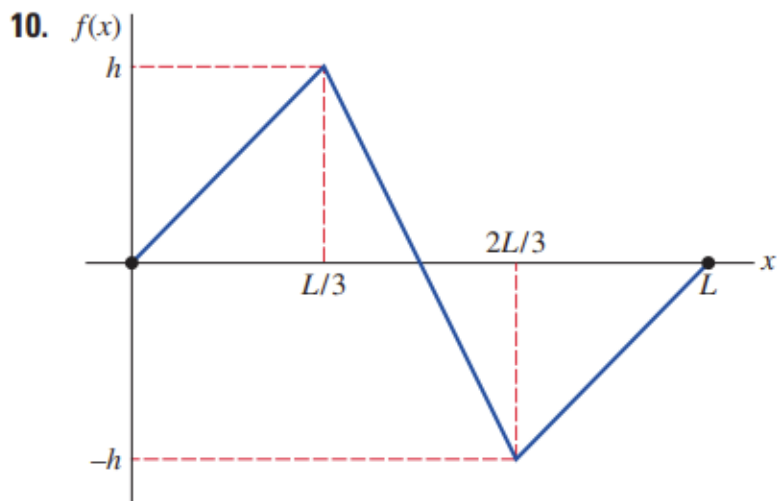
3. $u(0, t) = 0, \quad u(\pi, t) = 0, \quad t > 0$

$$u(x, 0) = 0, \quad \left. \frac{\partial u}{\partial t} \right|_{t=0} = \sin x, \quad 0 < x < \pi$$

5. $u(0, t) = 0, \quad u(1, t) = 0, \quad t > 0$

$$u(x, 0) = x(1 - x), \quad \left. \frac{\partial u}{\partial t} \right|_{t=0} = x(1 - x), \quad 0 < x < 1$$

In Problems 7–10, a string is tied to the x -axis at $x = 0$ and at $x = L$ and its initial displacement $u(x, 0) = f(x)$, $0 < x < L$, is shown in the figure. Find $u(x, t)$ if the string is released from rest.



Section 13.5 · ed. 5 p.709. p.710

In Problems 1–10, solve Laplace's equation (1) for a rectangular plate subject to the given boundary conditions.

4. $\frac{\partial u}{\partial x} \Big|_{x=0} = 0, \quad \frac{\partial u}{\partial x} \Big|_{x=a} = 0$

$u(x, 0) = x, \quad u(x, b) = 0$

10. $u(0, y) = 10y, \quad \frac{\partial u}{\partial x} \Big|_{x=1} = -1$

$u(x, 0) = 0, \quad u(x, 1) = 0$

In Problems 13 and 14, solve Laplace's equation (1) for a rectangular plate subject to the given boundary conditions.

13. $u(0, y) = 0, \quad u(a, y) = 0$

$u(x, 0) = f(x), \quad u(x, b) = g(x)$

Section 13.6 · ed. 5 p.716

In Problems 1 and 2, solve the heat equation $ku_{xx} = u_t$, $0 < x < 1$, $t > 0$ subject to the given conditions.

1. $u(0, t) = 100, \quad u(1, t) = 100$
 $u(x, 0) = 0$

6. Solve the boundary-value problem

$$k \frac{\partial^2 u}{\partial x^2} - hu = \frac{\partial u}{\partial t}, \quad 0 < x < \pi, \quad t > 0$$

$$u(0, t) = 0, \quad u(\pi, t) = u_0, \quad t > 0$$

$$u(x, 0) = 0, \quad 0 < x < \pi.$$

The PDE is a form of the heat equation when heat is lost by radiation from the lateral surface of a thin rod into a medium at temperature zero.

Section 15.3 · ed. 5 p.758

In Problems 1–6, find the Fourier integral representation of the given function.

$$2. f(x) = \begin{cases} 0, & x < \pi \\ 4, & \pi < x < 2\pi \\ 0, & x > 2\pi \end{cases}$$

In Problems 7–12, represent the given function by an appropriate cosine or sine integral.

$$10. f(x) = \begin{cases} x, & |x| < \pi \\ 0, & |x| > \pi \end{cases}$$